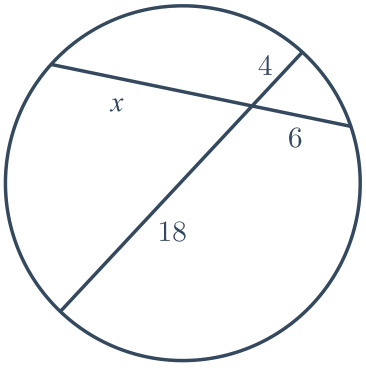
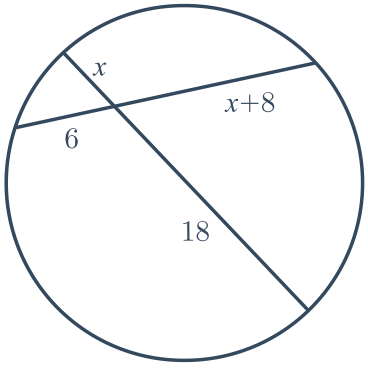
Lengths in chords, secants, and tangents

1 For each of the following diagrams, solve for x.

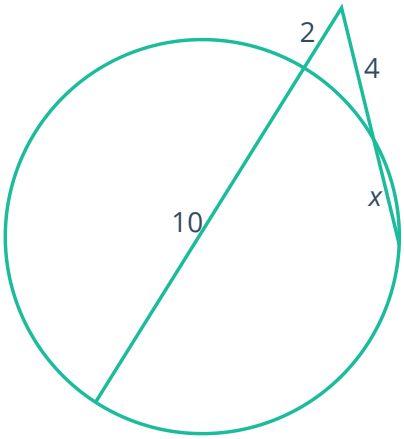
a



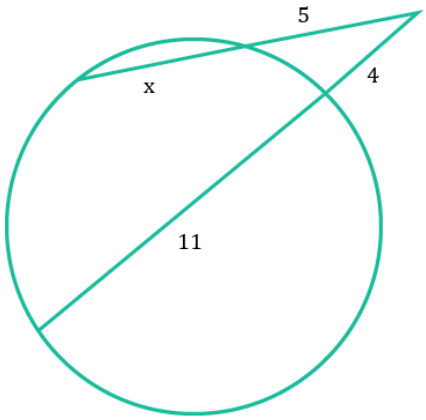
b



c

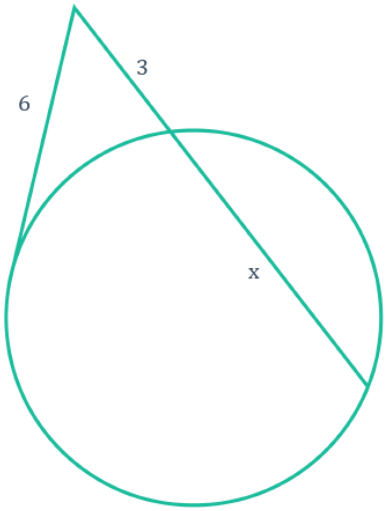


d

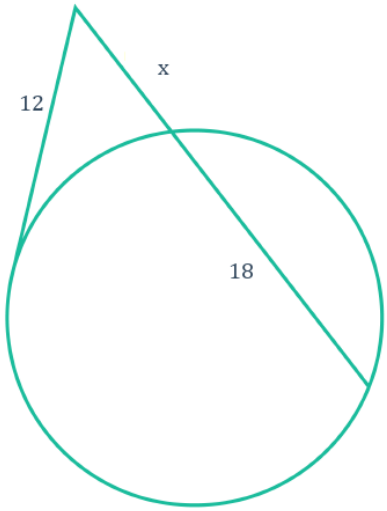


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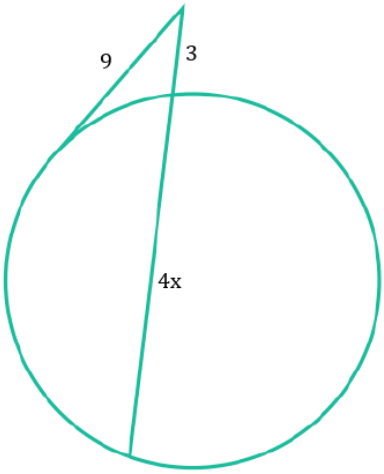
e



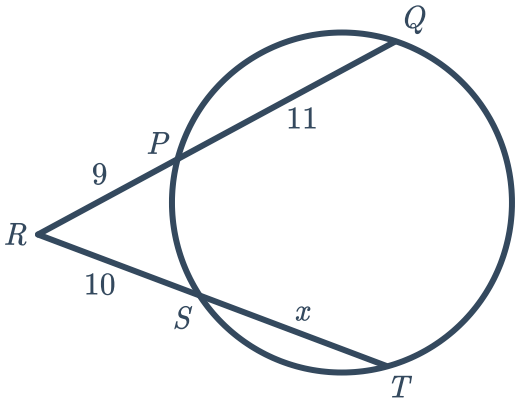
f



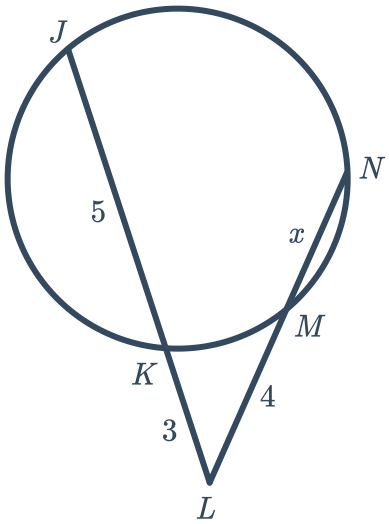
g



h



i



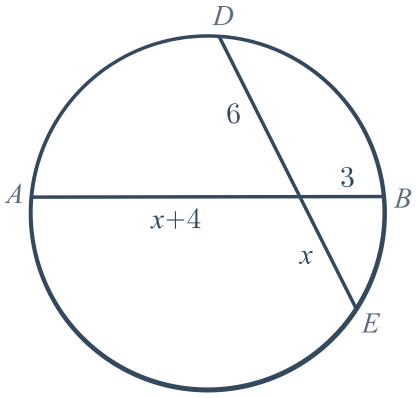
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2 Consider the circle with two intersecting chords.

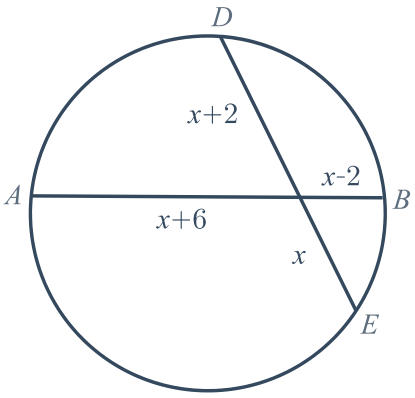
i Solve for x.

ii Find the length of the indicated chord.

a $\overline{AB}$

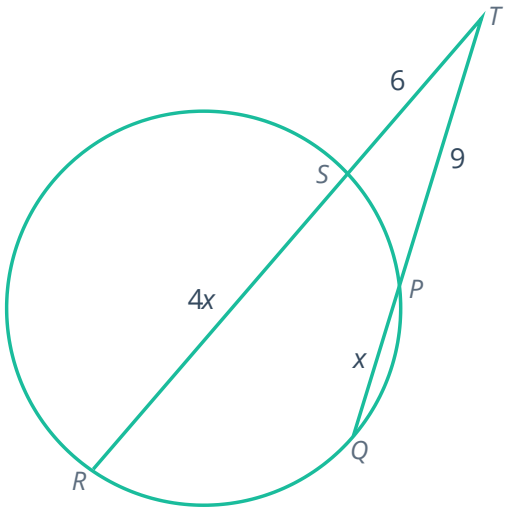


b $\overline{DE}$

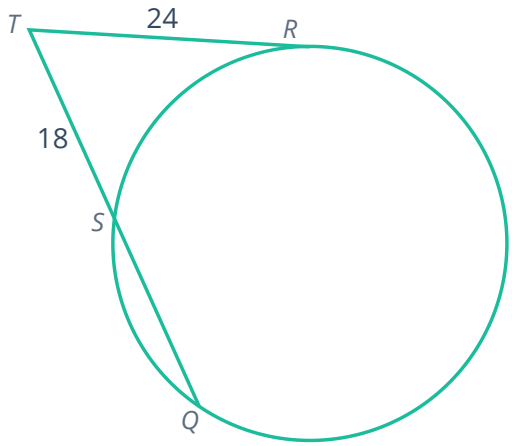


3 For each of the following, find the length of the indicated secant, tangent or chord.

a $\overline{RT}$

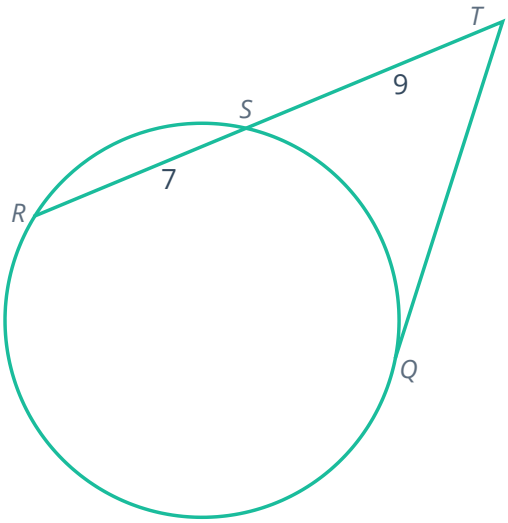


b $\overline{QS}$

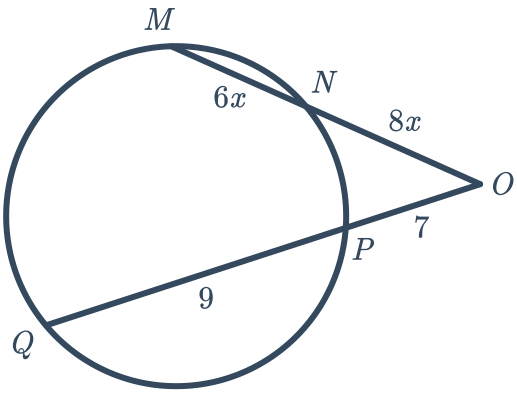


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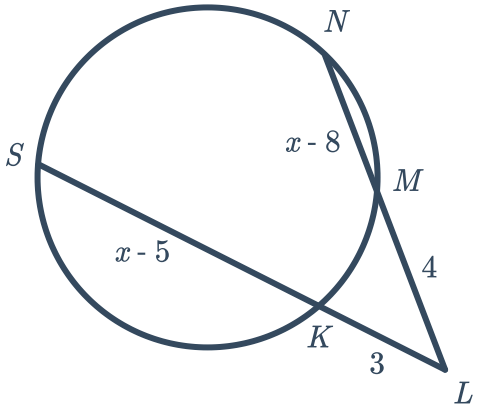
c $\overline{QT}$



d $\overline{MN}$



e $\overline{KS}$

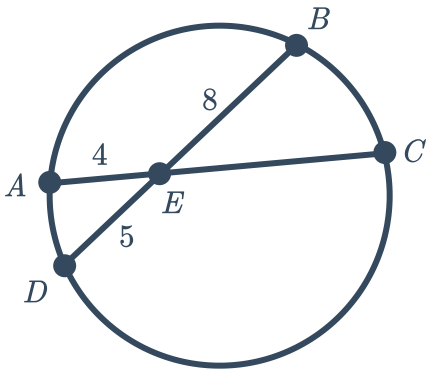


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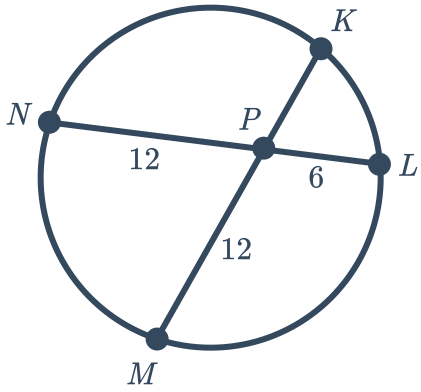
Additional questions

4

a Solve for EC.

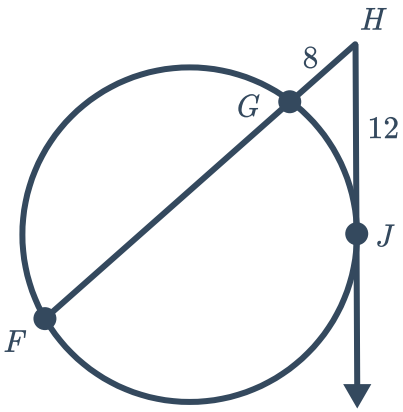


b Solve for PK.

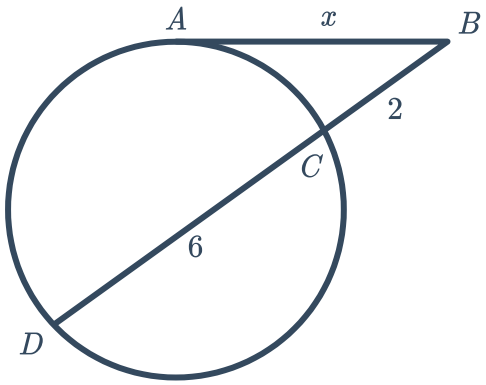


5

a Given $\overrightarrow{HJ}$ is tangent to the circle at J. Solve for GF.

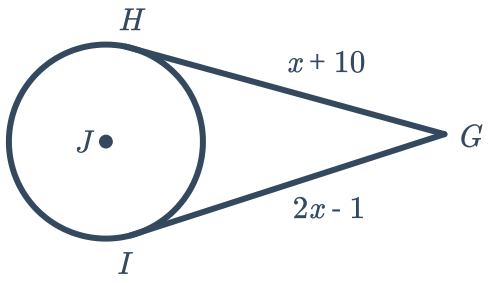


b Given $\overline{AB}$ is tangent to the circle at A. Solve for x.

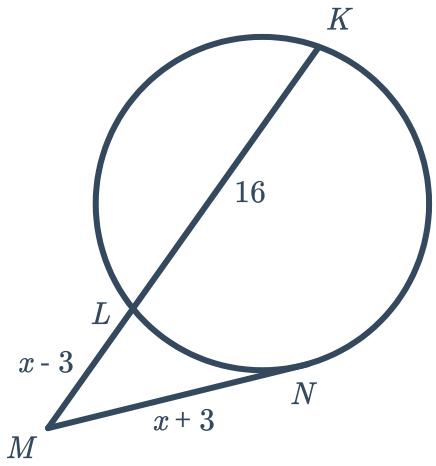


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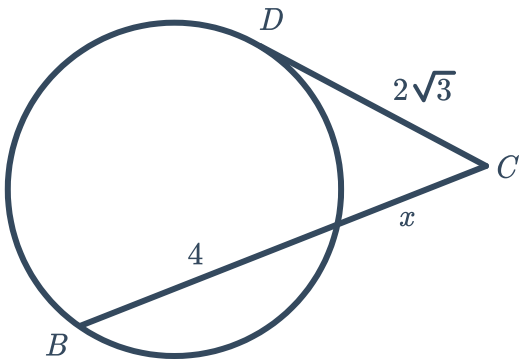
c Given: $\overline{HG}$ is tangent to circle J at H
$\overline{GI}$ is tangent to circle J at I
Solve for x.



d Given: $\overline{MN}$ is tangent to the circle at N.
Solve for ML.

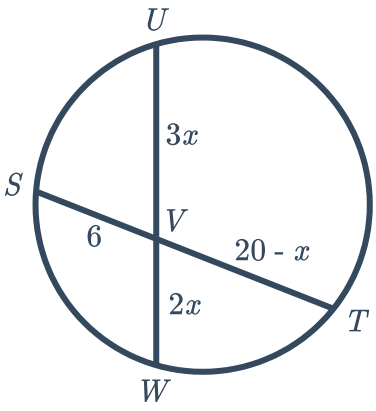


e Given: $\overline{DC}$ is tangent to the circle at D.
Solve for BC.

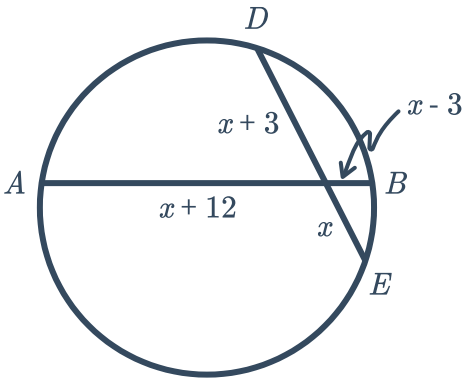


6

a Solve for UW.

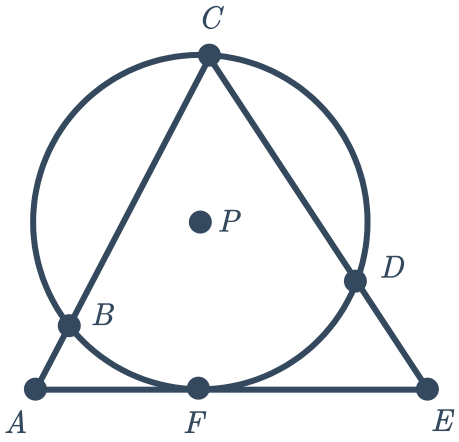


b Solve for DE.



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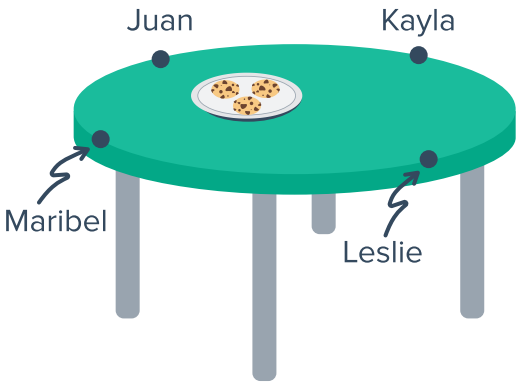
7 Given: $\overline{AE}$ is tangent to circle P at F
$AB = 2$, $BC = 9$, $FE = 12$, and $CD = 2(DE)$
Solve for the following:



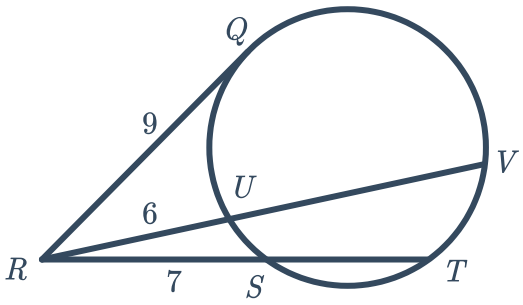
a AF

b DE

8 Juan, Kayla, Leslie, and Maribel are sitting around a table eating cookies. Maribel is 12 in. away from the middle of the cookie plate and Kayla is 26 in away from the middle of the plate. Juan is twice as far away from Leslie as he is from the middle of the cookie plate.
About how far away is Juan from the cookies?



9 Given: $\overline{RQ}$ is tangent to the circle at Q
About how much longer is $\overline{UV}$ than $\overline{ST}$?

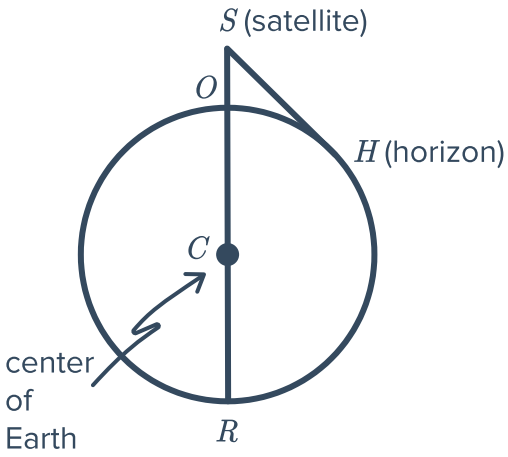


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10 A satellite is orbiting at point S, taking photos from above the earth's atmosphere. The horizon, point H, is the place where the earth appears to “meet the sky”.

a Describe $\overline{OS}$, $\overline{SH}$, and $\overline{OR}$ in terms of the scenario.

b The diameter of the earth is approximately $7920\ \text{mi}$. If the satellite has an altitude of $621\ \text{mi}$, about how far is the satellite from the earth's horizon?



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